PADOVA neuroscience CENTER

UNIVERSITÀ DEGLI STUDI DI PADOVA

Update - 19/10

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Supervisor: Professor Maurizio Corbetta

Point of Discussion

- **Period abroad**
  - Deco’s Lab → Nemesis, Contact with Jackub
- **Projects**
  - Nemesis
  - Awarenet?

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A Feature Space of Stroke Disease

PhD Emma Maria Sole Tosato

Supervisor: Professor Maurizio Corbetta

Rationale



Problem

In Stroke, we have several separated stories:

- Anatomical (**lesion**)
- Structural disconnections (**SDCs**)
- Functional features (**fMRI**)
- Physiological features (**EEG**)
- Phenotypic (demographics + clinical/behavioral **scores**)

→ different data types, different scales,



Idea

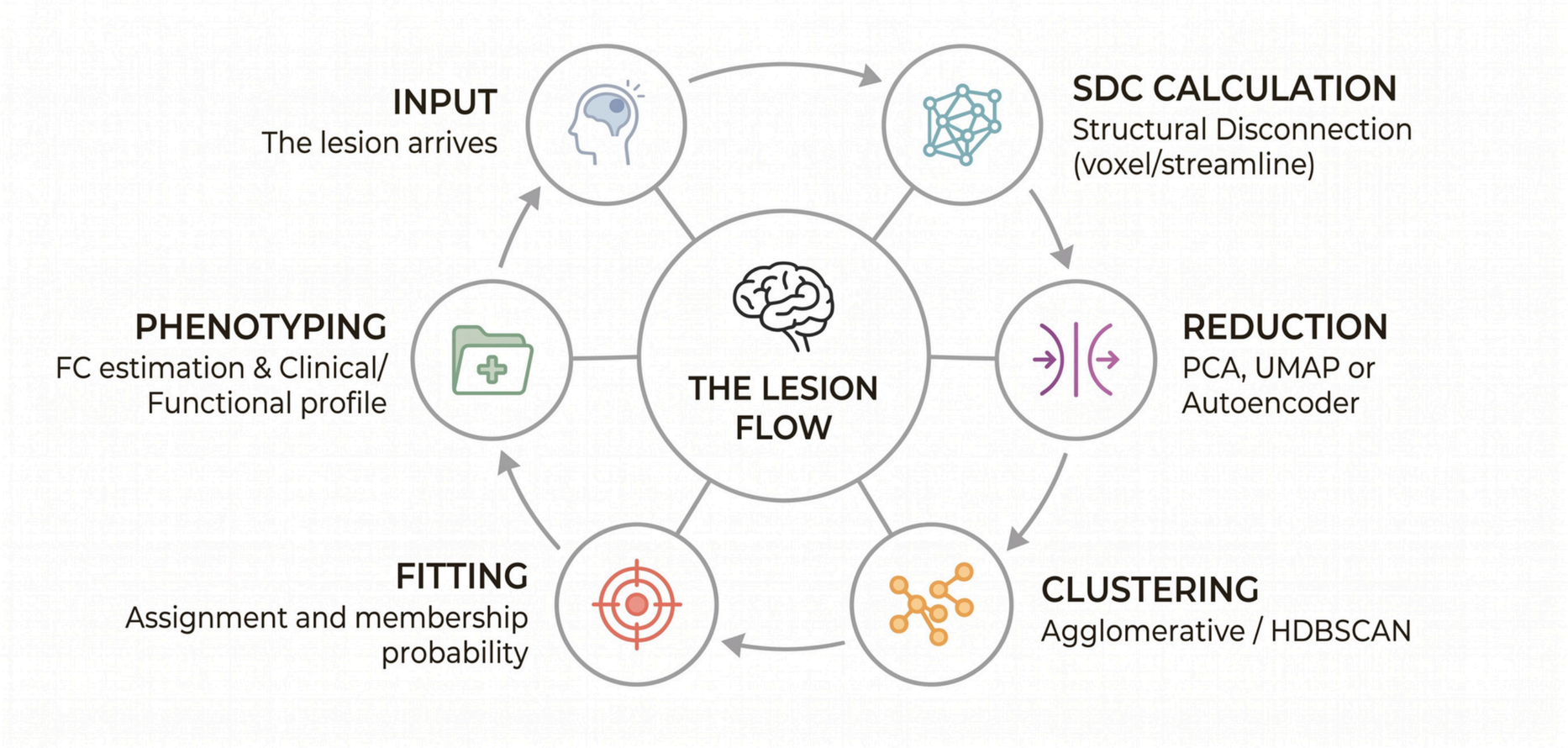
Identify **distinct patient profiles** per modality, in search of **canonical connectivity patterns** as targets for deficit-specific recovery interventions.



Hypothesis

Each modality has its own low-dimensional structure, whose clusters differ in behavioral profile and clinical.

Rationale


INPUT
The lesion arrives
SDC CALCULATION
Structural Disconnection
(voxel/streamline)
PHENOTYPING
FC estimation & Clinical/
Functional profile
THE LESION
FLOW
REDUCTION
PCA, UMAP or
Autoencoder
FITTING
Assignment and membership
probability
CLUSTERING
Agglomerative / HDBSCAN

Dataset

Dataset	Lesion mask	SDC
UCL-UK / UCLStrokeData	4119	4119
UKLFR / stroke_UKLFR	705	705
UKE / WAKEUP_acute	451	451
UNIPD / WashU	195	195
UNIPD / PSP	168	168
UNIPD / PASPORT	83	83
UNIPD / NEMESIS_T0	76	76
UKE / SFB936_ses01	56	56
Total	**5853**	**5853**

Dataset Counts

Group	**n**
Stroke (ST)	169
Healthy Controls (HC)	56
Total	**225**

UNIPD/WashU dataset FC matrices

Methods and Workflow

Data used: Lesion Masks

- MNI152 NLin6Asym, 8 datasets, 1 mm grid (182×218×182)
- 5,845 stroke patients (8 excluded)

Preprocessing Lesions

- Resample to 2 mm grid (nearest):
  - 91×109×91 = 902,629 voxels,
  - binarise at 0.5
- Out-of-brain voxels set to zero
- **Excluded (8):**
  - 3 empty masks,
  - 5 with >30% of the lesion outside the brain
- Constant voxels dropped

Per-subject measures

- **Volume**: number of lesion voxels × voxel size (0.008 ml at 2 mm)
- **Side**:
  - count lesion voxels left (L) and right (R) of the brain midline;
  - clinical label used when available; mask-based otherwise
- **Out-of-brain fraction**:
  - share of lesion voxels outside the brain mask, measured on the raw mask before zeroing;
  - used to decide exclusions (> 30%)
- **Location**:
  - each voxel assigned to 1 of 7 atlas-based categories; per patient, the lesion fraction in each;
  - dominant = largest

Methods and Workflow

Data used: Structural Disconnections (SDCs):

- **BCBToolKit**
  - takes each lesion mask and estimates which white-matter connections it interrupts, using tractography from healthy controls
  - Output: disconnection probability (0–1) per voxel, plus tract-level summaries
- Same 5,845 patients (same 8 exclusions)

Preprocessing SDC

- Resample to 2 mm grid (nearest): 91×109×91 = 902,629 voxels; probabilities kept continuous (no binarisation)
- Out-of-brain voxels set to zero (5,844 patients affected; median 0.8% of a map's mass, max 15.3%)
- Excluded (8): same as the lesions
- Constant voxels dropped

Per-subject measures

- Disconnection load: sum of the probabilities inside the brain

Methods and Workflow

Dimensionality reduction

Summarise each patient (221,955 lesion voxels, or 291,345 disconnection voxels) in a few coordinates

- **UMAP**
  - neighbour-graph embedding (2, 3, 10 dimensions)
  - Distance: Dice (binary lesions) or Euclidean (lesions, disconnection)
  - Grid over n_neighbors and min_dist; choice = stable plateau of trustworthiness and HDBSCAN probe
- **PCA**
  - centred, not standardised
  - computed on the 5,845 × 5,845 subject-by-subject matrix (voxels ≫ patients);
  - 2–150 components
- **PCA + varimax**
  - orthogonal rotation of the first k components (Thiebaut de Schotten et al. 2020)
  - centred, not standardised
  - 2–150 components
  - Same total variance (e.g. 42 components: 90.0%), no variance ordering

Workflow

	PCA	PCA + Varimax
Criterion	maximise variance	maximise simplicity (few large weights, the rest ≈ 0)
Voxel weights	spread over many voxels	concentrated on a few voxels
Order	by variance explained	none
Total variance (k components)	same	same, redistributed across components
Reads as	"a mix of everything"	"a region / a pattern"
Best for	input to clustering	interpreting components (maps)

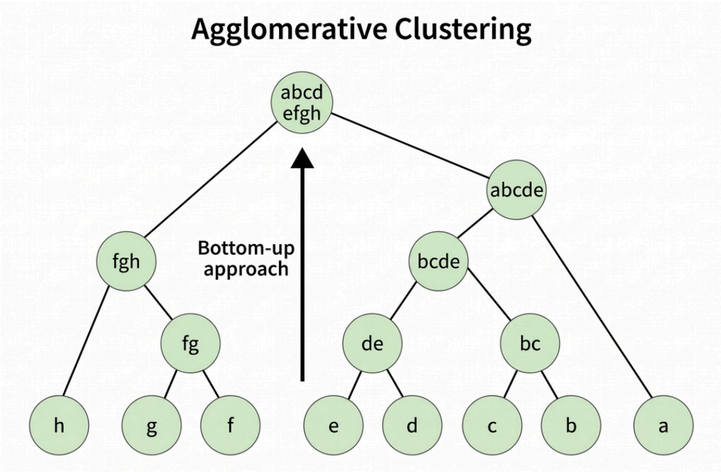
Workflow

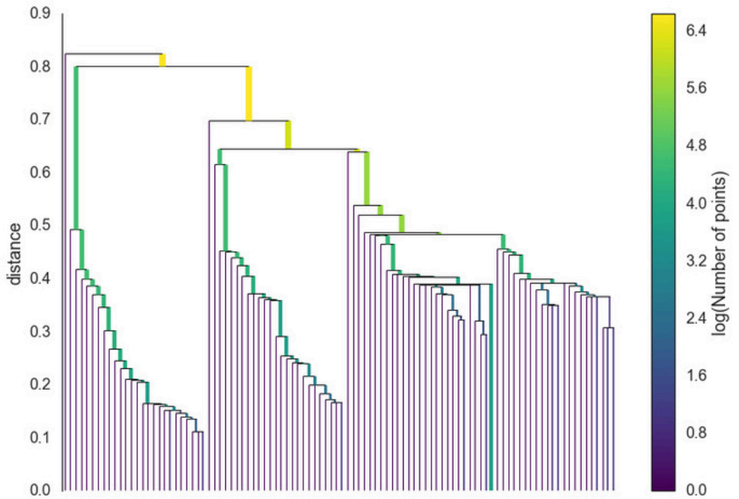
Clustering

Tuning: parameter grid per embedding, scored with internal indices (silhouette, Calinski–Harabasz, Davies–Bouldin, noise fraction)

Production: selected combinations re-run and saved

- **Agglomerative**: k = 2–30 × linkage × metric; k read from the indices, several k kept in parallel
- **HDBSCAN**:
  - min_cluster_size × min_samples; noise points left unassigned
  - eom and ma leaf





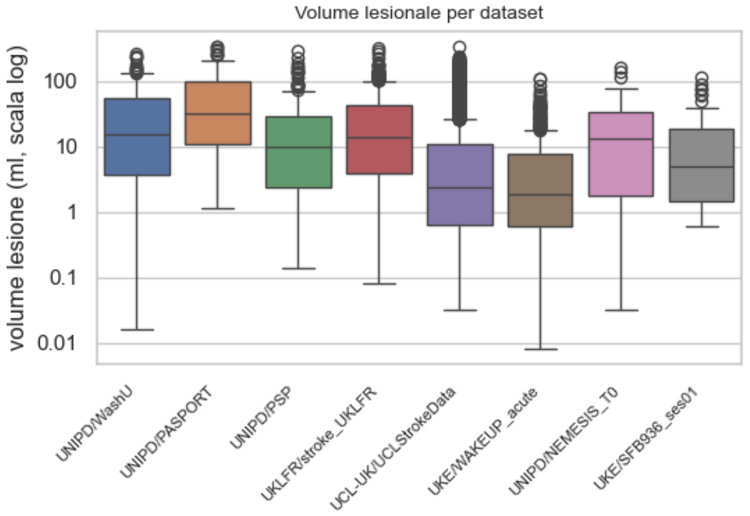
Cluster description

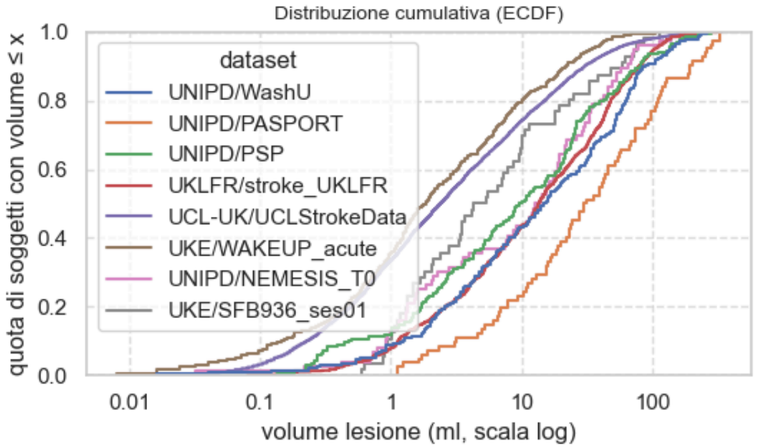
- Demographics: age, sex, education
- Clinical: NIHSS
- Lesion: volume, side, dominant location
- Descriptive statistics only (mean ± SD, median [IQR], %), with the number of patients who have the data

Results

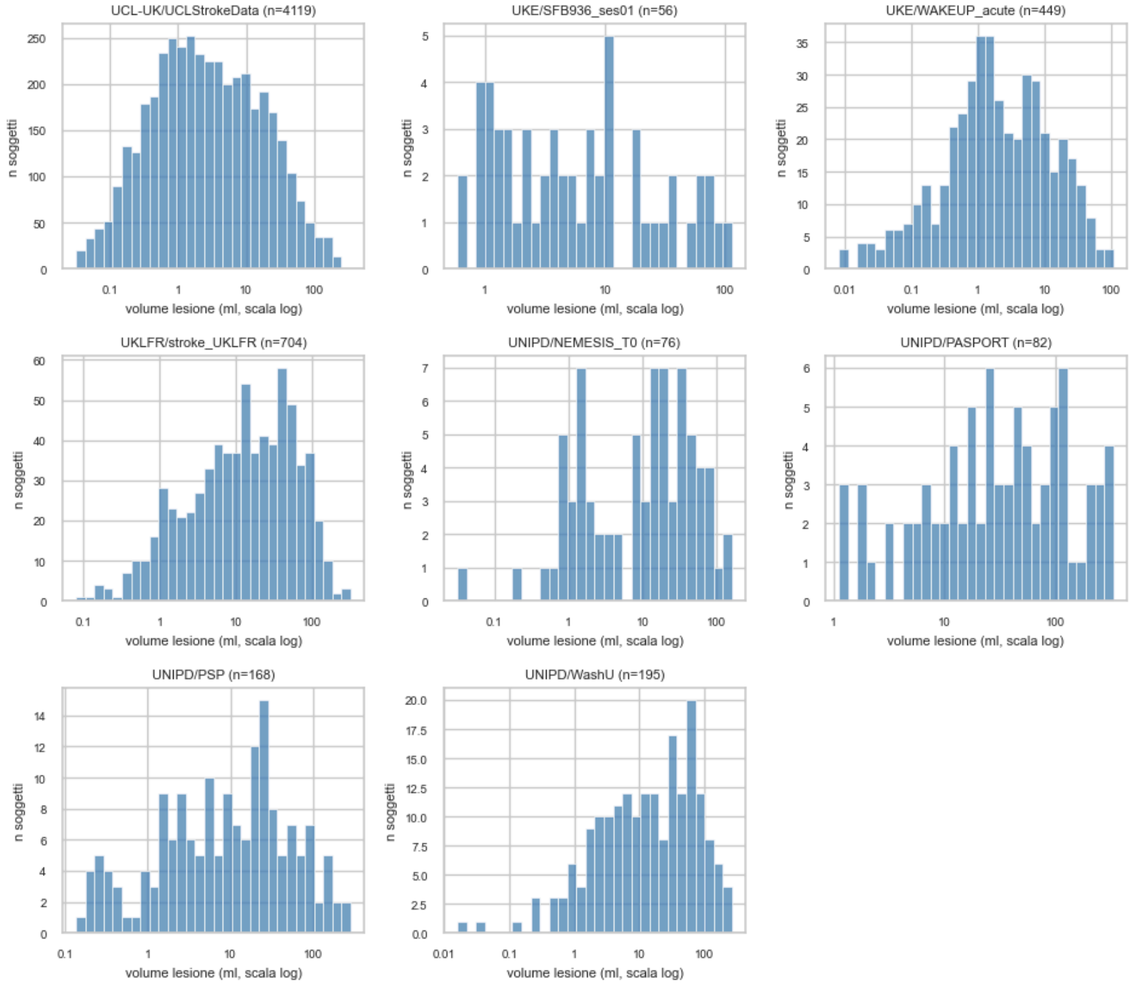
Preprocessing - Excluded subjects

- 8 patients excluded from both the lesion and SDC matrices (explicit list, reviewed one by one)
  - 3 empty masks: 1 truly empty, 2 only at 2 mm (8 and 3 voxels at 1 mm, lost in resampling)
  - 5 with 41–100% of the lesion outside the brain
- Small lesions are kept, unless they vanish at 2 mm
- Tract-wise matrix: 14 more patients dropped (13 UCL-UK, 1 WashU) because no tract is hit (all zeros); they remain in the other matrices

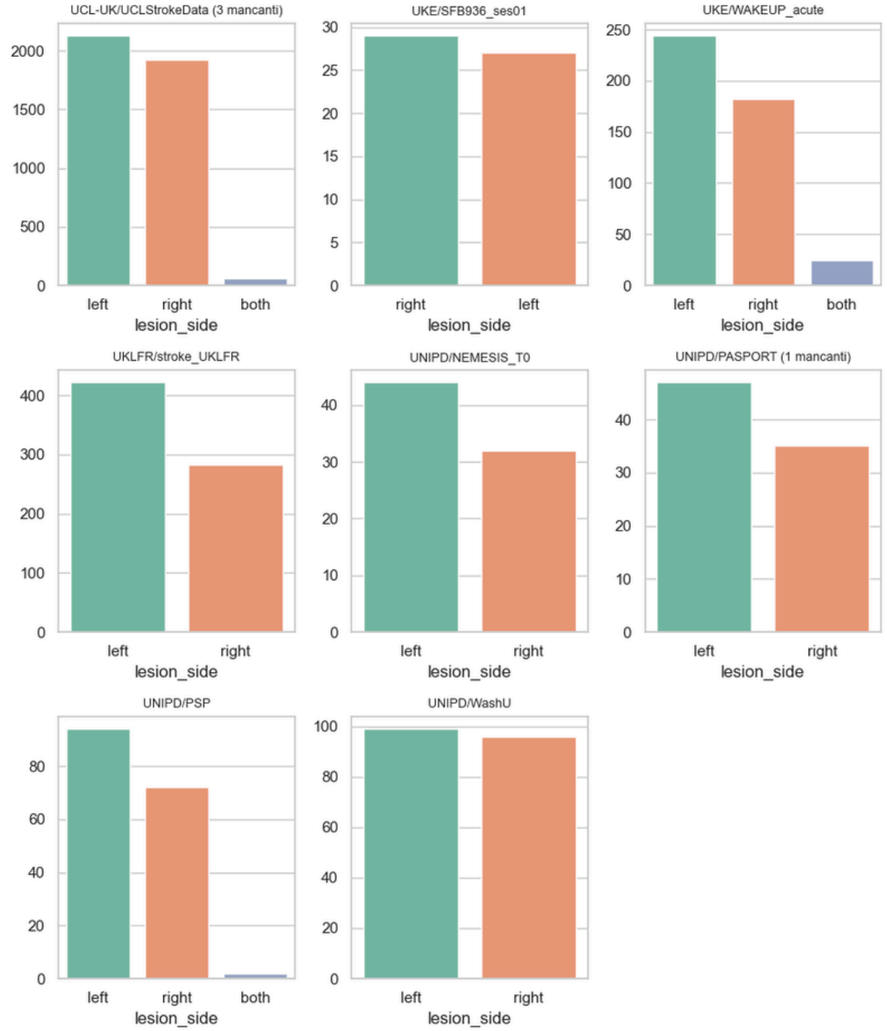




Results



Lesion Volume Per Dataset

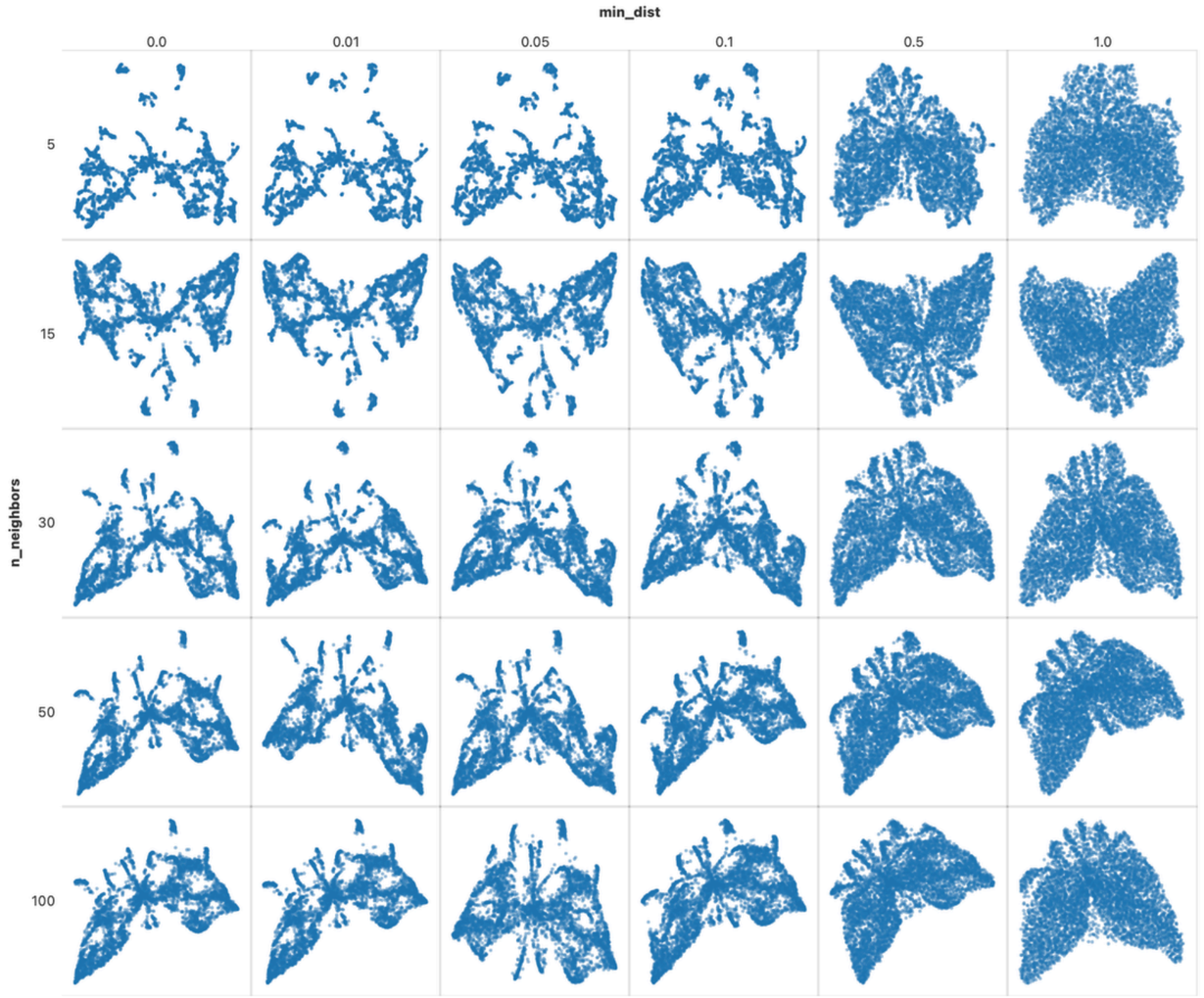


Lesion Side Per Dataset

Results

Dimensionality Reduction - UMAP

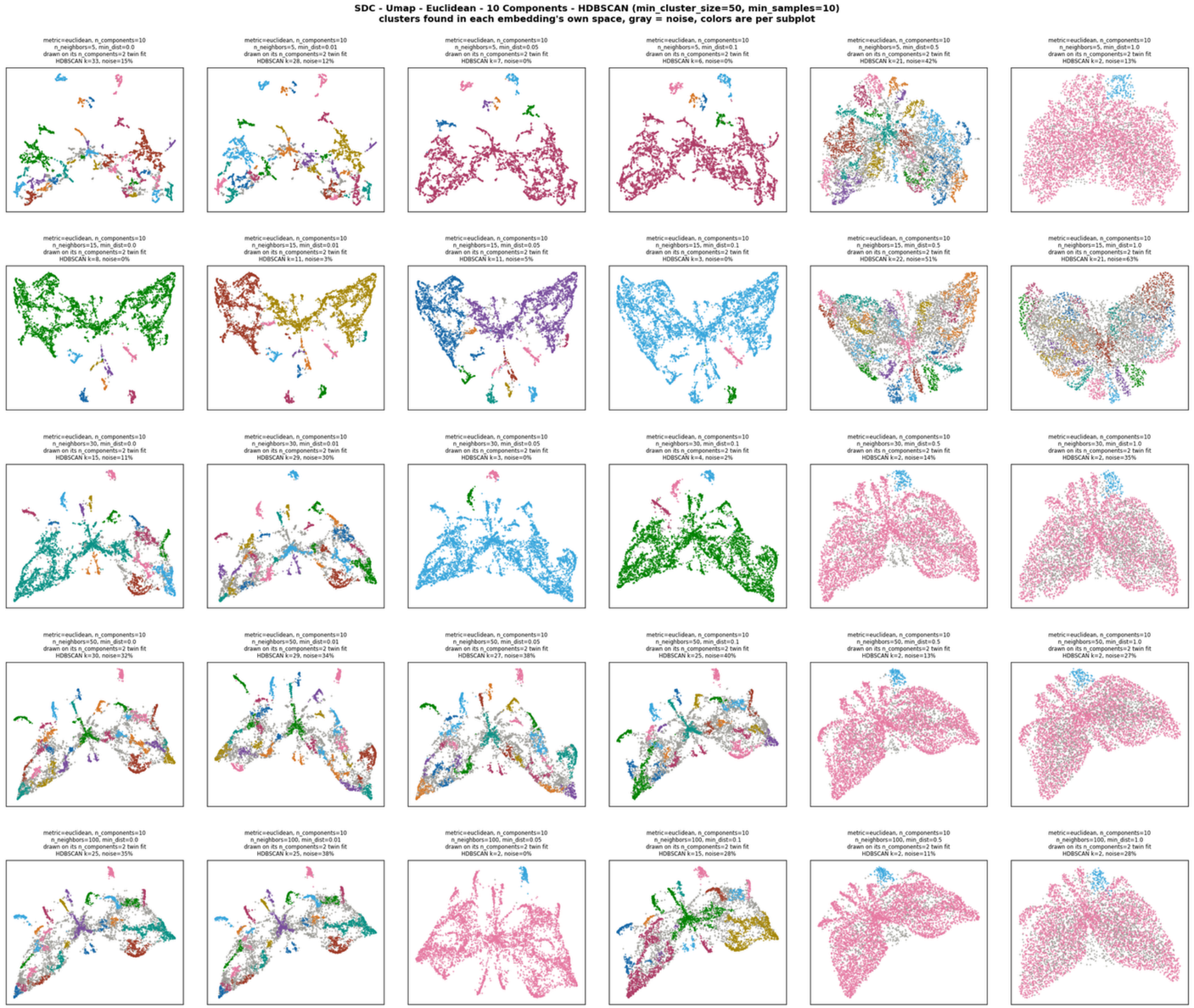
- Parameters chosen for n_components ∈ {2, 3, 10}.
  - n_neighbors=15,
  - min_dist=0.0
- min_dist=0.0: values ⩾ 0.5 wash out the structure
- HDBSCAN results stay stable
- At 5 the clusters fragment, and ⩾ 30–100 the clusters fragment



Results

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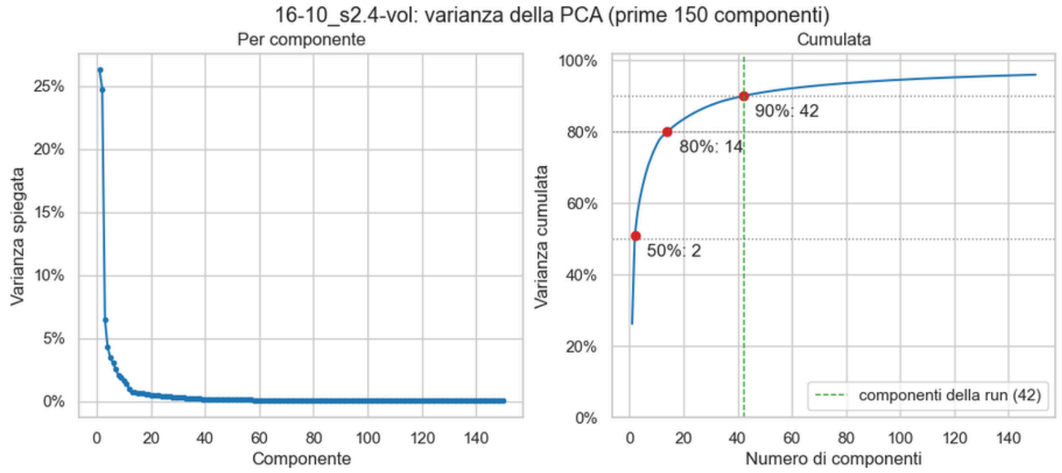


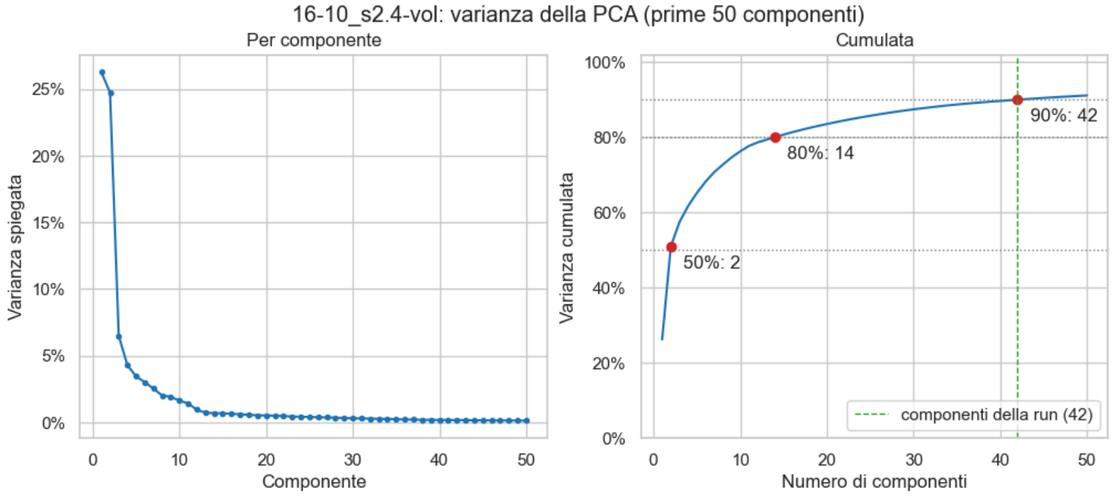
Results

Dimensionality Reduction - PCA

- 42% components is chosen by cumulative explained variance.
- The first two components dominate (26% + 25%) and are nearly equal in size.
- The scree plot shows a sharp drop after the second component and a smaller one after the 12th

Componente	Varianza
1	26.3%
2	24.7%
3	6.5%
4	4.3%
5	3.4%
10	1.6%
12	0.9%
15	0.7%





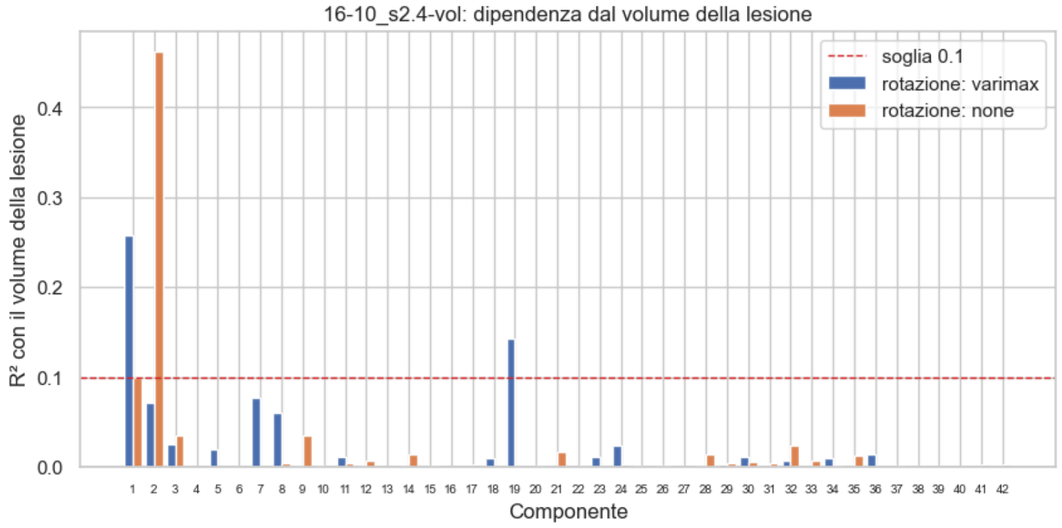
Results

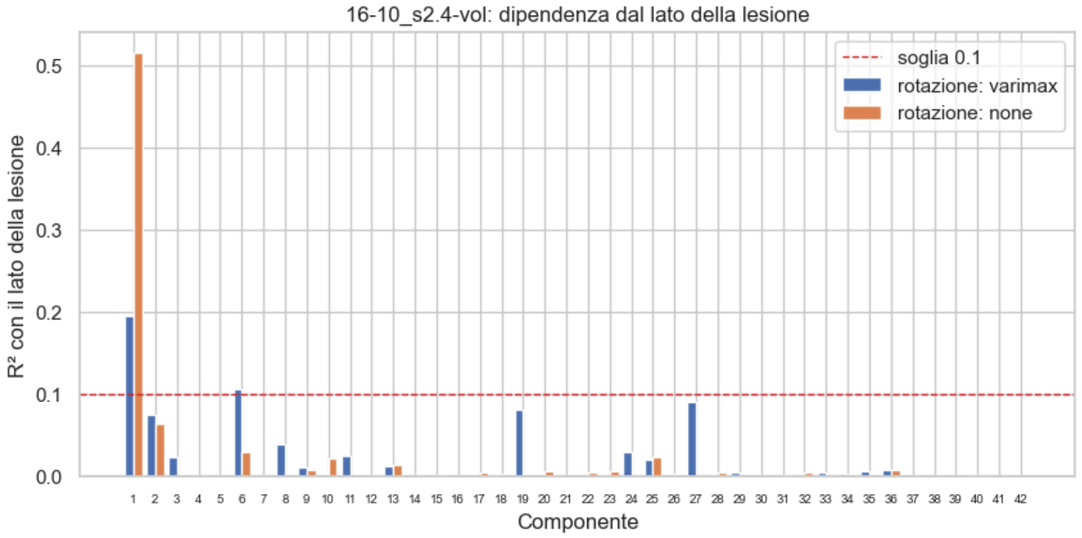
Dimensionality Reduction - PCA

- R^2 is the share of score variability explained by variables (volume and side)
  - R^2 = 0.04: component almost independent of volume.
  - R^2 = 0.50: component that mainly measures volume.
- 0 = the components say nothing about it; 1 = they explain it completely

	No rotation		Varimax	
k	volume	side	volume	side
1	0.10	0.52	0.26	0.20
2	0.56	0.58	0.33	0.27
3	0.60	0.58	0.47	0.35
5	0.60	0.58	0.53	0.48
7	0.60	0.61	0.61	0.59
9	0.64	0.62	0.65	0.61
15	0.66	0.67	0.68	0.65
30	0.72	0.72	0.75	0.73
42	0.78	0.75	0.78	0.75

R^2 with the volume and with the side of the lesion, using the first k components (in order of variance)

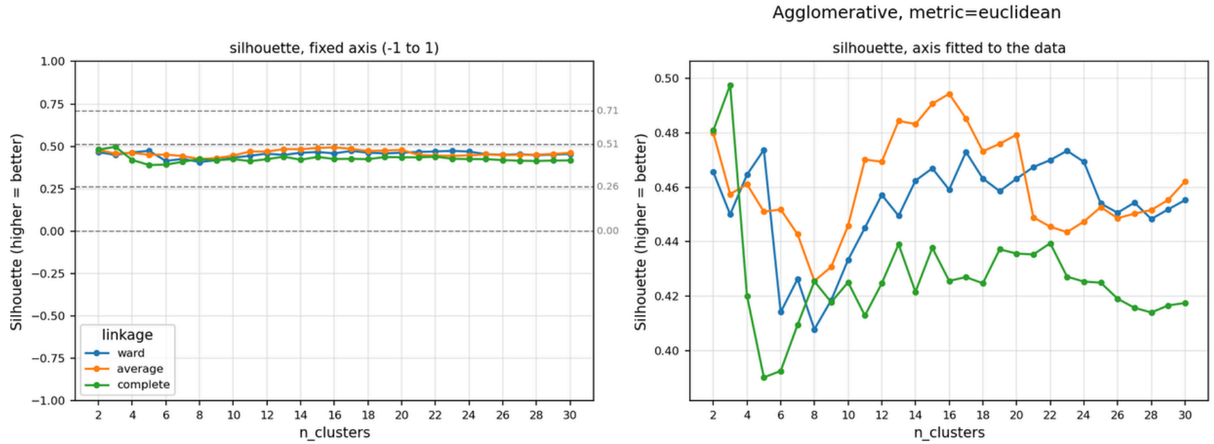




Results

Agglomerative Clustering on UMAP

- 2 umap components.
- Agglomerative Clustering
  - *Metrics:* euclidean, cosine, manhattan
  - *Linkage*: ward, average, complete



nc2				
Metric / linkage	Top 5 k (highest silhouette first)	**Range of top 5 k**	Silhouette, top 5 (6th)	Silhouette range over all k
euclidean / ward	5, 23, 17, 22, 24	**5–24**	0.469–0.474 (0.467)	0.408–0.474
euclidean / average	16, 15, 17, 13, 14	**13–17**	0.483–0.494 (0.480)	0.426–0.494
euclidean / complete	3, 2, 22, 13, 15	**2–22**	0.438–0.498 (0.437)	0.390–0.498
cosine / average	2, 3, 7, 8, 6	**2–8**	0.700–0.832 (0.697)	0.594–0.832
cosine / complete	2, 3, 4, 19, 10	**2–19**	0.685–0.788 (0.679)	0.547–0.788
manhattan / average	3, 15, 16, 13, 14	**3–16**	0.478–0.496 (0.472)	0.406–0.496
manhattan / complete	3, 19, 21, 2, 20	**2–21**	0.447–0.488 (0.447)	0.334–0.488

Results

HDBSCAN Clustering on UMAP

- 2 umap components.
- HDBSCAN Clustering
  - sklearn clustering
  - soft clustering

nc2, leaf (from 0.470 to 0.665)

Rank	Min cluster size / min samples	Silhouette	Noise	Clusters found	Largest cluster share	Mean membership	Membership < 0.5
1	10 / 20	0.665	48.6%	84	2.3%	0.55	51%
2	15 / 20	0.662	47.7%	78	2.3%	0.61	44%
3	20 / 20	0.659	46.2%	70	2.3%	0.65	39%
4	10 / 10	0.647	41.1%	144	1.3%	0.58	49%
5	30 / 20	0.634	44.4%	51	2.8%	0.68	34%
6th	15 / 10	0.620	39.5%	113	1.3%	0.63	42%

HDBSCAN Soft Clustering

nc2

Rank	Min cluster size / min samples	Silhouette	Noise	Clusters found	Largest cluster share
1	10 / 20	0.609	35.3%	73	3.0%
2	20 / 20	0.604	33.2%	59	3.9%
3	15 / 20	0.600	34.5%	64	3.9%
4	10 / 10	0.553	29.3%	122	3.0%
5	15 / 5	0.552	20.8%	112	2.8%
6th	20 / 5	0.544	20.7%	98	2.8%

HDBSCAN Clustering

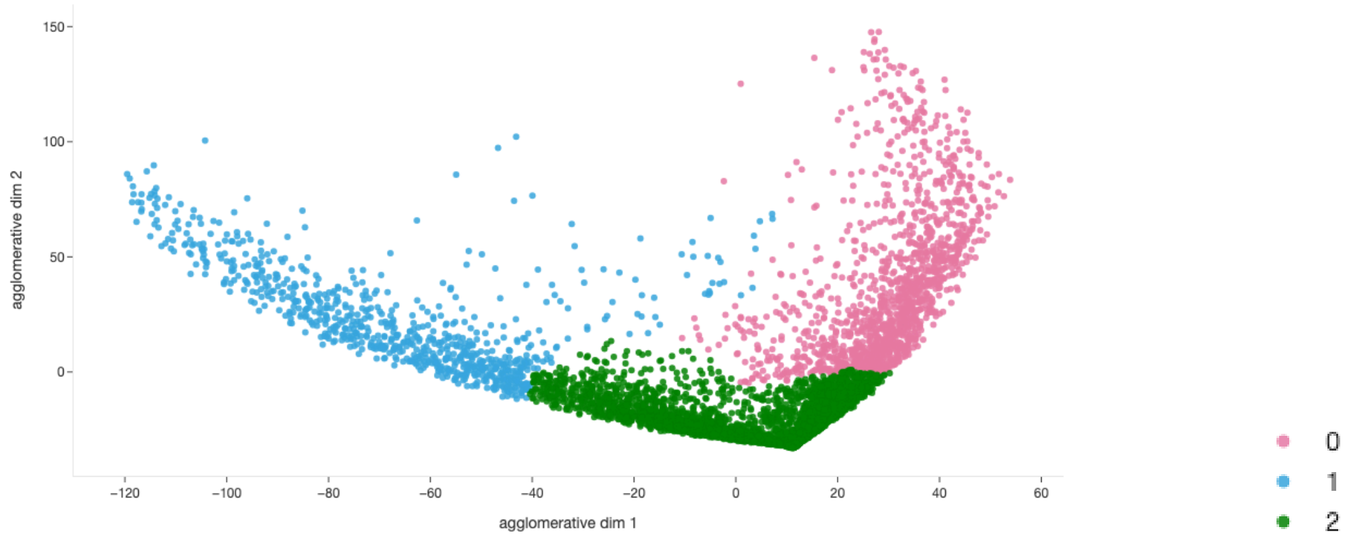
Results

Agglomerative Clustering on PCA

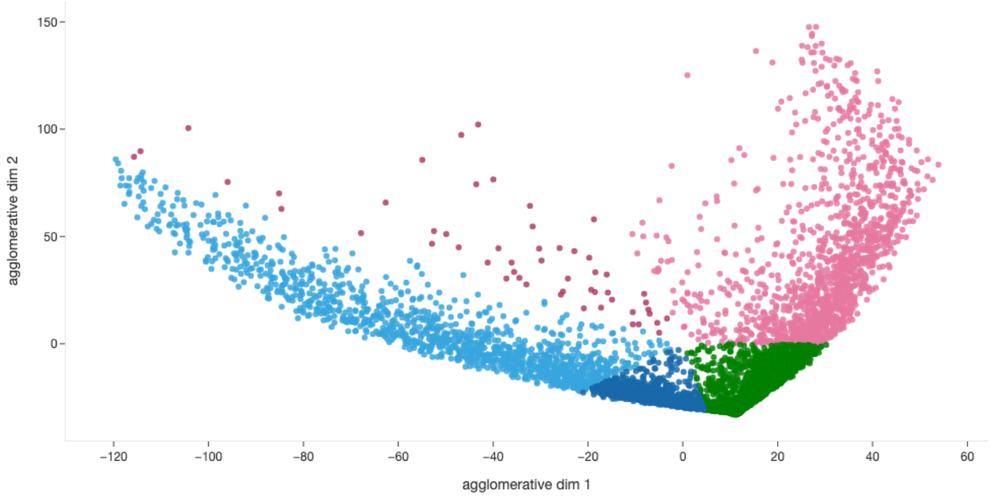
- 2 PCA components

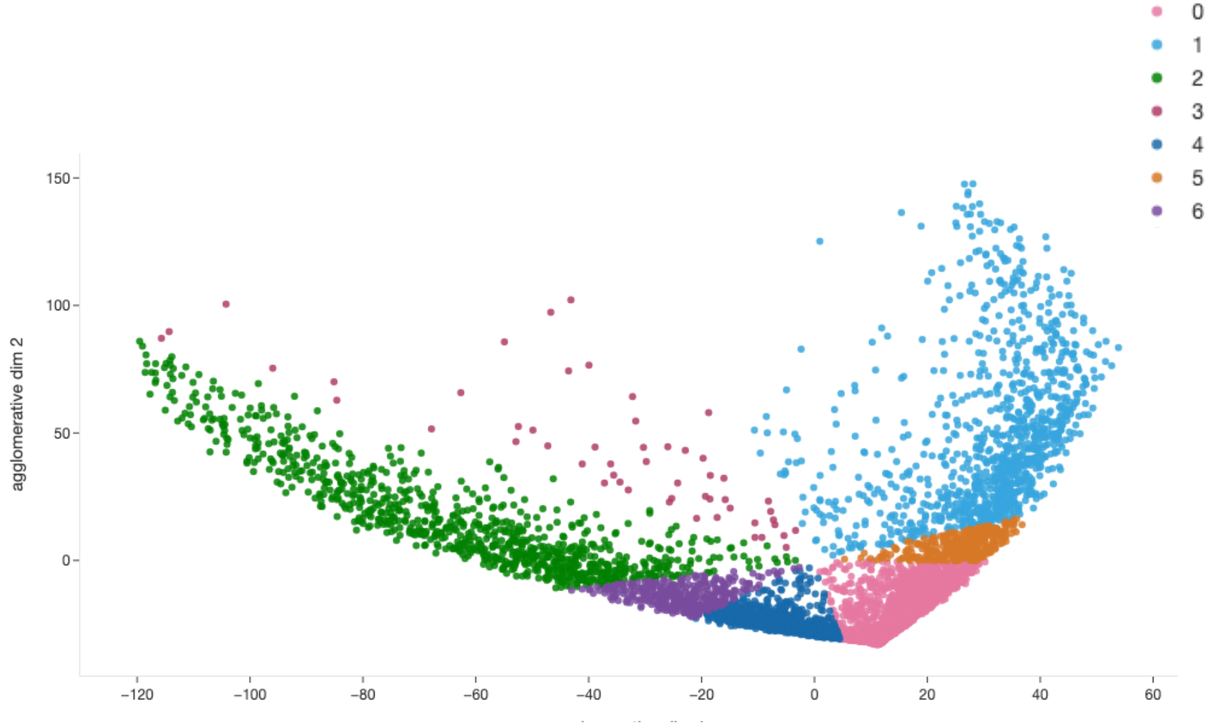
PCA, nc2				
Metric / linkage	Top 5 k (highest silhouette first)	**Range of top 5 k**	Silhouette, top 5 (6th)	Silhouette range over all k
euclidean / ward	3, 2, 4, 6, 5	**2–6**	0.456–0.536 (0.452)	0.362–0.536
euclidean / average	2, 3, 4, 5, 9	**2–9**	0.451–0.544 (0.449)	0.371–0.544
euclidean / complete	3, 4, 5, 2, 6	**2–6**	0.482–0.577 (0.420)	0.338–0.577
cosine / average	3, 4, 18, 13, 24	**3–24**	0.634–0.739 (0.633)	0.505–0.739
cosine / complete	3, 17, 18, 15, 19	**3–19**	0.637–0.729 (0.637)	0.481–0.729
manhattan / average	2, 3, 4, 7, 8	**2–8**	0.480–0.570 (0.471)	0.339–0.570
manhattan / complete	2, 3, 4, 6, 7	**2–7**	0.484–0.563 (0.462)	0.344–0.563

Agglomerative Clustering - Non rotated components



Results


150
100
50
0
agglomerative dim 2
−120
−100
−80
−60
−40
−20
0
20
40
60
agglomerative dim 1
0
1
2
3
4


150
100
50
0
agglomerative dim 2
−120
−100
−80
−60
−40
−20
0
20
40
60
0
1
2
3
4
5
6

Results

HDBSCAN Clustering on PCA

- 2 PCA components
- PCA non rotated
- PCA Varimax

PCA, nc2, eom (from −0.048 to 0.644; 27 of 28 combinations with clusters)

Rank	Min cluster size / min samples	Silhouette	Noise	Clusters found	Largest cluster share	Mean membership	Share with membership < 0.5
1	50 / 10	0.644	79.8%	11	6.9%	0.14	95%
2	20 / 10	0.578	74.8%	29	2.1%	0.09	97%
3	15 / 10	0.565	73.4%	39	2.0%	0.07	98%
4	10 / 10	0.541	71.2%	54	2.0%	0.05	99%
5	30 / 5	0.533	67.8%	29	3.5%	0.08	97%
6th	75 / 10	0.500	34.6%	2	63.4%	0.57	36%

Membership = soft probability of belonging to the assigned cluster; mean and share computed on non-noise patients.

PCA, nc2, leaf (from 0.378 to 0.676; 27 of 28 combinations with clusters)

Rank	Min cluster size / min samples	Silhouette	Noise	Clusters found	Largest cluster share	Mean membership	Share with membership < 0.5
1	15 / 20	0.676	84.5%	22	1.7%	0.10	97%
2	10 / 20	0.653	83.5%	31	1.7%	0.07	98%
3	30 / 10	0.649	79.4%	21	2.0%	0.11	96%
4	50 / 10	0.644	79.8%	11	6.9%	0.14	95%
5	30 / 20	0.632	83.6%	10	6.8%	0.15	93%
6th	20 / 20	0.631	82.7%	17	4.9%	0.11	96%

Membership = soft probability of belonging to the assigned cluster; mean and share computed on non-noise patients.

HDBSCAN Soft Clustering